

USER MANUAL

Manhattan Associates Payment Simulation System

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1. Introduction

This user manual provides clear, simple instructions for navigating and operating the Manhattan Associates Payment Simulation System.

The system demonstrates a complete e-commerce payment workflow:

- Creating and authorizing an order
- Submitting warehouse settlements (including partial & full settlements)
- Viewing all orders and transaction history

The manual is organized into short steps and visual examples to make demonstrations smooth and intuitive.

2. Starting the System

The system runs as two local services: **Backend (Express + SQLite)** and **Frontend (React)**.

Step 1: Download the files from Github

The Front End & Server Files can be downloaded from our group repository:

[FelixBote/IS-Capstone-Group-10](#)

The files needed to run the system are in the “Main Project Folder”

Step 2: Install Node Js

Download and install Node Js using the default installation options:

[Node.js — Download Node.js®](#)

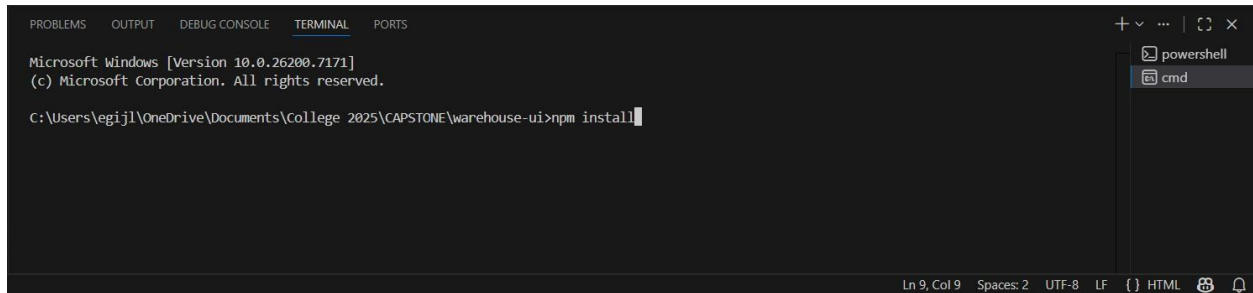
Step 3: Install the Node Modules in both the front end & Server folders

In a terminal:

```
cd server
npm install
```

```
Cd frond-end
npm install
```

SCREENSHOT: npm install typed into a terminal

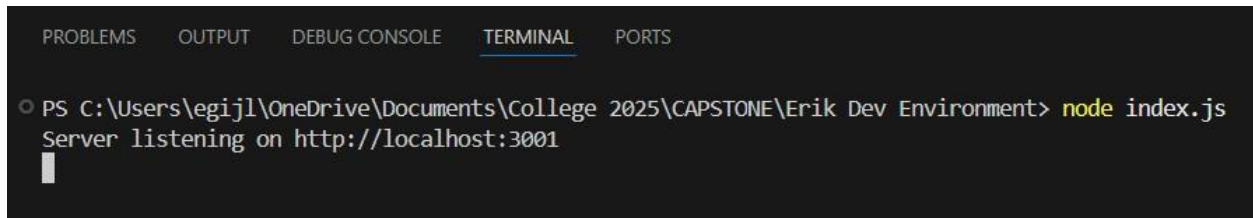


Step 4: Start the Backend

In a terminal:

```
cd server
node index.js
```

SCREENSHOT: Backend terminal running (“Server listening on port 3001”)



Step 5: Start the Frontend

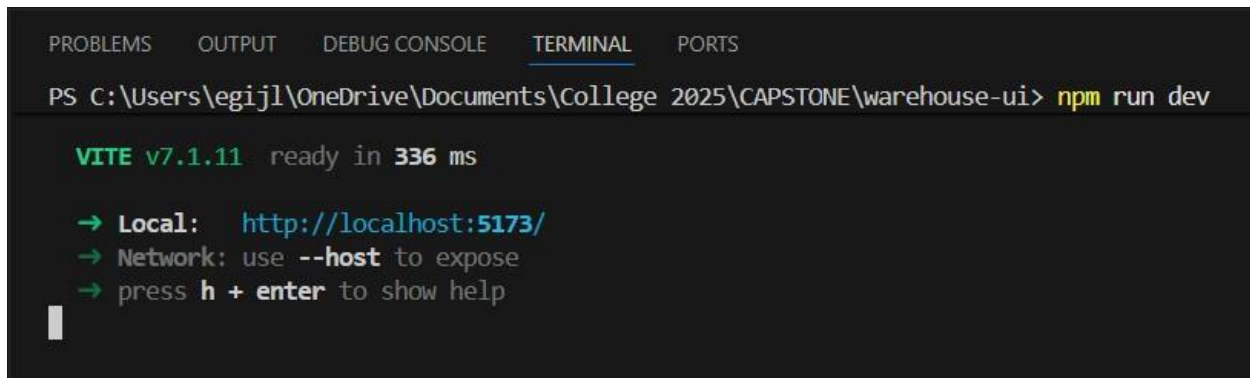
In another terminal:

```
cd frontend
npm run dev
```

Visit:

<http://localhost:5173>

SCREENSHOT: Starting the Front-End User Interfaces

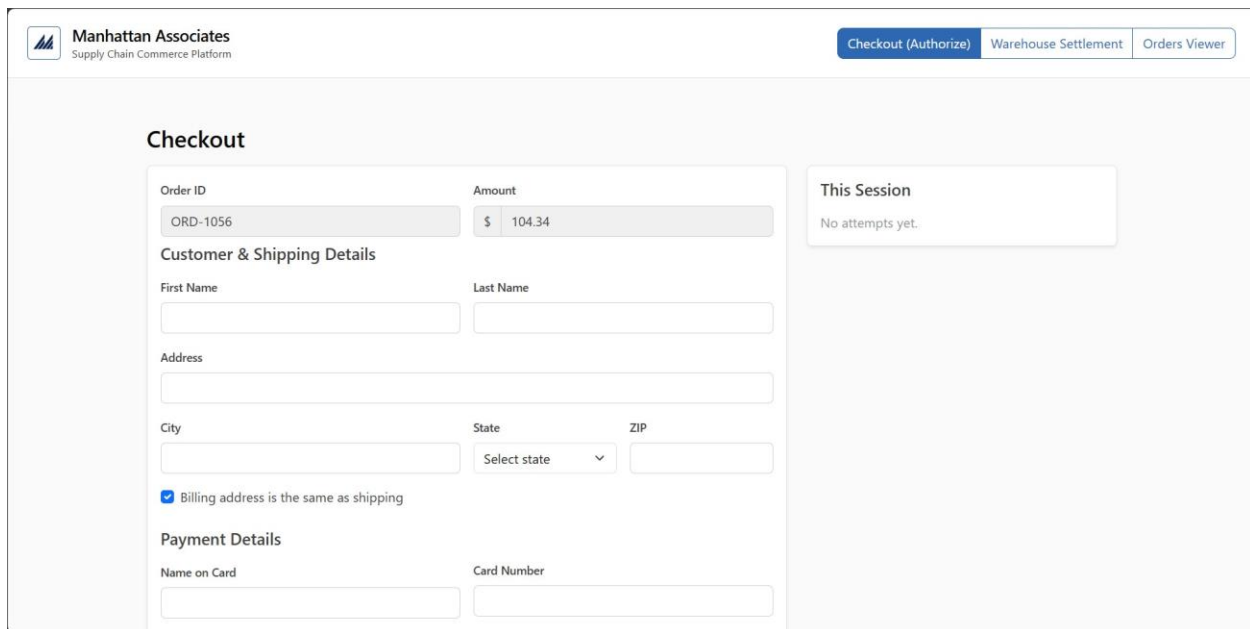


```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\egijl\OneDrive\Documents\College 2025\CAPSTONE\warehouse-ui> npm run dev

VITE v7.1.11 ready in 336 ms

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
```

SCREENSHOT: Browser showing main page at localhost:5173:



The screenshot shows the checkout page of the Manhattan Associates Supply Chain Commerce Platform. The page has a header with the company logo and name, and three navigation buttons: "Checkout (Authorize)", "Warehouse Settlement", and "Orders Viewer". The main content area is titled "Checkout" and contains several sections:

- Order ID:** A text box containing "ORD-1056".
- Amount:** A text box showing "\$ 104.34".
- Customer & Shipping Details:**
 - First Name:** A text box.
 - Last Name:** A text box.
 - Address:** A text box.
 - City:** A text box.
 - State:** A dropdown menu with "Select state" and a downward arrow.
 - ZIP:** A text box.
 - ☒ Billing address is the same as shipping
- Payment Details:**
 - Name on Card:** A text box.
 - Card Number:** A text box.

On the right side of the checkout form, there is a box titled "This Session" with the text "No attempts yet."

3. Checkout / Payment Processing

This interface simulates the experience of a customer placing an order and submitting credit card information.

3.1 Opening the Checkout Page

Navigate to:

Checkout (Authorize):

Checkout (Authorize)	Warehouse Settlement	Orders Viewer
----------------------	----------------------	---------------

or simply open the site's landing page.

SCREENSHOT: Blank Checkout page (Order ID + Amount already filled)

Manhattan Associates
Supply Chain Commerce Platform

Checkout (Authorize) Warehouse Settlement Orders Viewer

Checkout

Order ID	Amount
ORD-1056	\$ 104.34

Customer & Shipping Details

First Name	Last Name	
Address		
City	State	ZIP
	Select state	

☒ Billing address is the same as shipping

Payment Details

Name on Card	Card Number

This Session
No attempts yet.

3.2 Enter Customer Information

Fill in:

- First Name
- Last Name
- Address
- City
- ZIP Code

SCREENSHOT: Checkout page with customer info filled in

Checkout

Order ID	Amount
ORD-1056	\$ 104.34

Customer & Shipping Details

First Name	Last Name
Test	User

Address

1111 Main St

City	State	ZIP
Woodstock	Georgia	30188

☒ Billing address is the same as shipping

3.3 Enter Payment Information

Enter:

- Card Number (Luhn validated + auto-formatted)
- Expiration (MM/YY)
- CVV (brand-aware: 3 or 4 digits)
- Cardholder Name

SCREENSHOT: Card fields with formatted number + brand indicator

Payment Details

Name on Card	Card Number	
Test User	4111 1111 1111 1111	
Expiry (MM/YY)	CVV	
12/29	...	

3.4 Submit the Payment

Click **Pay**.

System performs:

- Full client-side validation
- Luhn checksum
- Masking rules
- Error highlighting if fields are incomplete

The system then contacts the Beeceptor mock provider.

3.5 View Authorization Results

Possible outcomes:

- **Approved**
- **Insufficient Funds**
- **Incorrect Details**
- **Server Error**

SCREENSHOT: Success banner

Checkout

Payment authorized

Order ORD-1056 authorized for \$104.34 (Card **** 1111).

Order ID

ORD-1057

Amount

\$ 268.48

Customer & Shipping Details

First Name

Last Name

This Session

ORD-1056

12/3/2025, 10:45:43 AM

Amount: \$104.34

SUCCESS

SCREENSHOT: Insufficient Funds banner

Checkout

Insufficient funds

The issuing bank declined the charge due to insufficient funds.

Order ID

ORD-1056

Amount

\$ 104.34

Customer & Shipping Details

First Name

Test

Last Name

User

This Session

ORD-1056

12/3/2025, 10:45:03 AM

Amount: \$104.34

INSUFFICIENT FUNDS

SCREENSHOT: Incorrect Details banner

Checkout

Incorrect card details

One or more card fields appear invalid. Please double-check and retry.

Order ID

ORD-1062

Amount

\$ 352.58

Customer & Shipping Details

First Name

Last Name

This Session

ORD-1062

12/3/2025, 10:46:38 AM

Amount: \$352.58

INCORRECT DETAILS

SCREENSHOT: Server Error banner

Checkout

Payment provider error

There was a temporary problem with the payment service. Please try again.

Order ID

ORD-1070

Amount

\$ 409.87

Customer & Shipping Details

First Name

Last Name

This Session

ORD-1070

12/3/2025, 10:48:20 AM

Amount: \$409.87

PROVIDER_ERROR

Once a payment is **Approved**, the order becomes available for settlement.

4. Warehouse Settlement UI

This interface allows staff to finalize orders by submitting shipment amounts.

Navigate to:

Warehouse Settlement:



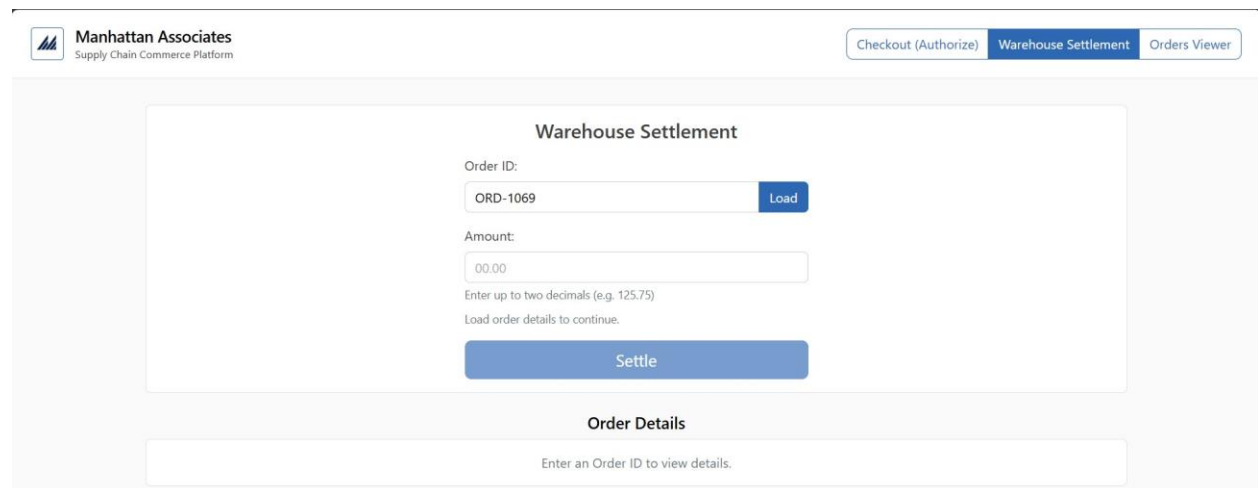
A horizontal navigation bar with three buttons: 'Checkout (Authorize)', 'Warehouse Settlement' (highlighted in blue), and 'Orders Viewer'.

4.1 Load an Order

Enter a valid **Order ID** from the Checkout UI.

Click **Load Order**.

SCREENSHOT: Settlement page with Order ID field filled



A screenshot of the Warehouse Settlement page. The top navigation bar shows 'Checkout (Authorize)', 'Warehouse Settlement' (active), and 'Orders Viewer'. The main content area is titled 'Warehouse Settlement' and contains the following fields and buttons:

- Order ID:** A text input field containing 'ORD-1069' and a blue 'Load' button.
- Amount:** A text input field containing '00.00'.
- Instructions:** 'Enter up to two decimals (e.g. 125.75)' and 'Load order details to continue.'
- Settle:** A large blue button.

Below the 'Warehouse Settlement' section is an 'Order Details' section with a text input field and the instruction 'Enter an Order ID to view details.'

SCREENSHOT: Loaded order details (status + authorized amount + last4)

The screenshot displays a web interface for settling an order. At the top, there is a 'Settle Order' section with a text input field containing 'ORD-1069' and a blue 'Load' button. Below this, a 'Recent:' section shows 'ORD-1069' in a grey box. An 'Amount:' section features a text input field with '00.00' and a note 'Enter up to two decimals (e.g. 125.75)'. A large blue 'Settle' button is positioned below these fields. The main section is titled 'Order Details' and contains a table with the following information:

Order ID	Status
ORD-1069	AUTHORIZED
Card (last 4) **** * 1111	
Authorized Amount \$433.86	
Available to Settle \$433.86	
Settlements No settlements yet.	

If the order is not AUTHORIZED, the system will display an error.

4.2 Perform a Partial Settlement (Example)

Example:

Authorized amount = **\$433.86**

Enter:

25.00

Click **Settle Order**.

Expected result:

- Settlement succeeds
- Remaining amount = **\$408.86**
- Order stays in **AUTHORIZED** status

SCREENSHOT: Partial settlement success banner

Settlement recorded. Remaining \$408.86.

Warehouse Settlement

Order ID:

ORD-1069 Load

Recent: ORD-1069

Amount:

00.00

Enter up to two decimals (e.g. 125.75)

Settle

SCREENSHOT: Updated settlement history panel

Order ID	Status
ORD-1069	AUTHORIZED
Card (last 4) **** * 1111	
Authorized Amount \$433.86	
Available to Settle \$408.86	
Settlements	
12/3/2025, 3:52:46 PM	\$25.00

4.3 Perform the Final Settlement

Now enter the remaining amount:

105.02

Click **Settle Order**.

Expected result:

- Order becomes **SETTLED**
- Remaining amount = **0.00**

SCREENSHOT: Final settlement success banner

Order ORD-1069 fully settled.

Warehouse Settlement

Order ID:

ORD-1069

Load

Recent:

ORD-1069

Amount:

00.00

Enter up to two decimals (e.g. 125.75)

Nothing left to settle for this order.

Settle

SCREENSHOT: Settlement history showing two entries

Order ID	Status
ORD-1069	SETTLED
Card (last 4)	
**** * 1111	
Authorized Amount	
\$433.86	
Available to Settle	
\$0.00	
Settlements	
12/3/2025, 3:52:46 PM	\$25.00
12/3/2025, 4:04:08 PM	\$408.86

SCREENSHOT: Status updated to SETTLED

Order ID	Status
ORD-1069	SETTLED

5. Order History UI

Navigate to:

Orders Viewer

Checkout (Authorize)

Warehouse Settlement

Orders Viewer

This interface is used for reporting and reviewing full transaction history.

5.1 Navigating Tabs

Two tabs:

- **Current Orders** (PENDING + AUTHORIZED)
- **Past Orders** (SETTLED + ERROR)

SCREENSHOT: Orders page – Current Orders tab

Orders
View current and past orders with their payment status.

Current 46

Past 24

From
mm/dd/yyyy

To
mm/dd/yyyy

Search by order, customer, or status

Export CSV

Order #	Customer	Date ▼	Total	Status	Payment
ORD-1068	Test User	12/3/2025, 3:48:17 PM	\$428.13	AUTHORIZED	**** 1111
ORD-1067	Test User	12/3/2025, 3:48:17 PM	\$166.39	AUTHORIZED	**** 1111
ORD-1066	Test User	12/3/2025, 3:48:16 PM	\$172.82	AUTHORIZED	**** 1111
ORD-1065	Test User	12/3/2025, 3:48:16 PM	\$461.14	AUTHORIZED	**** 1111
ORD-1064	Test User	12/3/2025, 3:48:15 PM	\$213.47	AUTHORIZED	**** 1111
ORD-1063	Test User	12/3/2025, 3:48:14 PM	\$368.39	AUTHORIZED	**** 1111
ORD-1062	Test User	12/3/2025, 3:46:37 PM	\$352.58	AUTHORIZED	**** 1111
ORD-1061	Test User	12/3/2025, 3:46:36 PM	\$446.34	AUTHORIZED	**** 1111
ORD-1060	Test User	12/3/2025, 3:46:34 PM	\$371.87	AUTHORIZED	**** 1111

SCREENSHOT: Orders page – Past Orders tab

Orders
View current and past orders with their payment status.

Current 46

Past 24

From
mm/dd/yyyy

To
mm/dd/yyyy

Status
All

Search by order, customer, or status

Export CSV

Order #	Customer	Date ▼	Total	Status	Payment
ORD-1070	Test User	12/3/2025, 3:48:19 PM	\$409.87	ERROR	 1111
ORD-1069	Test User	12/3/2025, 3:48:18 PM	\$433.86	SETTLED	 1111
ORD-1055	Erik Gijlstra	12/3/2025, 3:36:00 PM	\$449.61	ERROR	 4444
ORD-1054	Felix Botero	12/3/2025, 2:32:28 PM	\$356.76	SETTLED	 1111
ORD-1052	Felix Botero	11/20/2025, 10:14:06 PM	\$208.50	SETTLED	 0010
ORD-1051	Felix Botero	11/20/2025, 10:13:50 PM	\$302.44	SETTLED	 0010
ORD-1050	Test User	11/20/2025, 10:06:20 PM	\$188.23	SETTLED	 1111
ORD-1049	Test User	11/20/2025, 10:02:20 PM	\$295.19	SETTLED	 1111
ORD-1048	Test User	11/20/2025, 9:56:59 PM	\$73.37	SETTLED	 1111

5.2 Sorting Orders

Click column headers to sort by:

- Order #
- Customer
- Date
- Total
- Status

SCREENSHOT: A column header in “sorted” state

Order #	Customer ▲	Date	Total	Status	Payment
ORD-1055	Erik Gijlstra	12/3/2025, 3:36:00 PM	\$449.61	ERROR	 4444
ORD-1041	Erik Gijlstra	11/10/2025, 3:29:10 PM	\$354.14	SETTLED	 4444
ORD-1054	Felix Botero	12/3/2025, 2:32:28 PM	\$356.76	SETTLED	 1111
ORD-1052	Felix Botero	11/20/2025, 10:14:06 PM	\$208.50	SETTLED	 0010
ORD-1051	Felix Botero	11/20/2025, 10:13:50 PM	\$302.44	SETTLED	 0010

5.3 Filtering Orders

Use:

- Date range filter
- Status filter (Current Orders tab)
- Search bar (order ID / customer / status)

SCREENSHOT: Filters being used

The screenshot shows the 'Orders' section with the following filters applied:

- Date Range:** From 11/01/2025 to 11/30/2025.
- Status:** SETTLED (indicated by a red arrow and a blue box).
- Search:** John (indicated by a red arrow).

The table below shows the filtered results:

Order #	Customer ▲	Date	Total	Status	Payment
ORD-1031	John Smith	11/7/2025, 9:38:34 PM	\$122.15	SETTLED	 7158
ORD-1028	John Smith	11/7/2025, 9:16:19 PM	\$199.18	SETTLED	 1111

Showing 1-2 of 2 orders

Rows per page: 20 (dropdown menu)

Navigation: Previous, Page 1 of 1, Next

5.4 Opening the Order Details Panel

Click any row to expand full details:

- Order info
- Authorization results
- Settlement list
- Amount remaining

SCREENSHOT: Expanded detail panel

Order Details #ORD-1041

Close

Order Customer: Erik Gijlstra Total: \$354.14 Status: SETTLED Created: 11/10/2025, 3:29:10 PM	Authorization Outcome: SUCCESS Amount: \$354.14 Token: STATIC_TOKEN_ORD-1041	Settlements Available to settle: \$0.00 Amount: \$20.00 11/10/2025, 3:29:30 PM SUCCESS Amount: \$334.14 11/10/2025, 3:29:36 PM SUCCESS
---	---	--

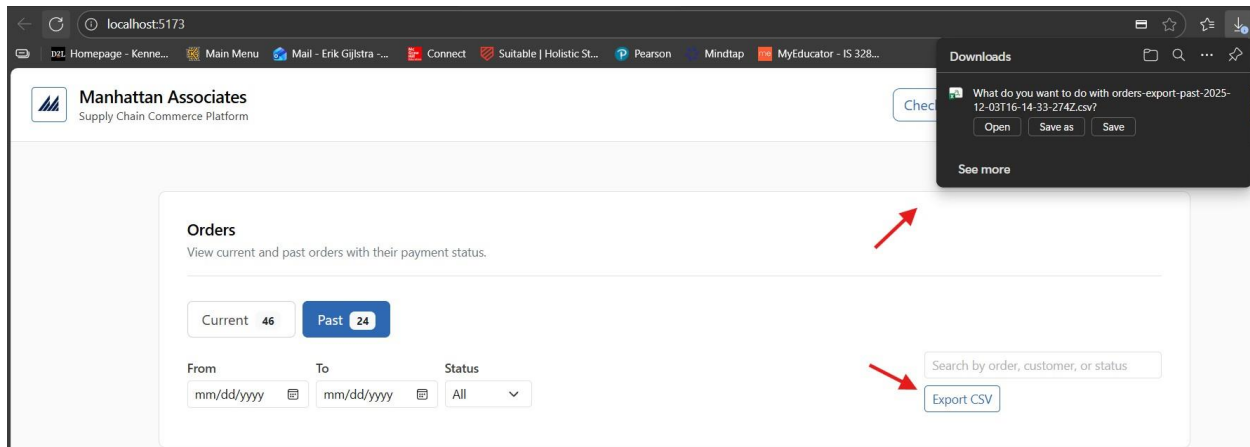
Order #	Customer ▲	Date	Total	Status	Payment
ORD-1055	Erik Gijlstra	12/3/2025, 3:36:00 PM	\$449.61	ERROR	**** 4444
ORD-1041	Erik Gijlstra	11/10/2025, 3:29:10 PM	\$354.14	SETTLED	**** 4444
ORD-1054	Felix Botero	12/3/2025, 2:32:28 PM	\$356.76	SETTLED	**** 1111

5.5 Exporting Orders to CSV

Click the **Export CSV** button.

Exports the current filtered/sorted results — not just the visible page.

SCREENSHOT: CSV export button



6. Resetting the System

If you need a clean demo:

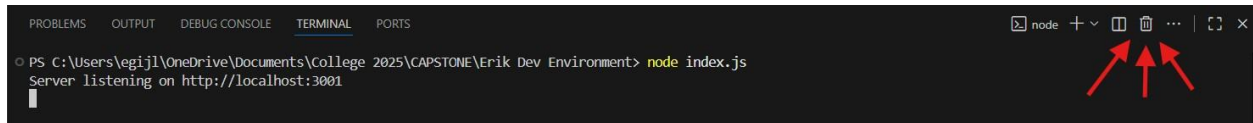
1. Close Webpage
2. Stop the backend
3. Restart backend
4. Restart frontend (if needed)

The system should start fresh.

SCREENSHOT: Close User Interface Tab



SCREENSHOT: Close Backend Terminal



7. Troubleshooting Guide

Issue	Cause	Fix
Frontend cannot connect	Backend not running	Start backend first
Authorization always fails	Beeceptor offline	Wait 1–2 minutes and retry
No new orders appearing	SQLite file locked	Restart backend
Duplicate test data	DB not reset	Delete database.db
“Order Not Found” in settlement	Invalid Order ID	Use an AUTHORIZED order

8. Frequently Asked Questions (FAQ)

Q1: What card numbers should I use for testing?

Any standard-format test numbers work. Examples include:

- **Visa:** 4111 1111 1111 1111
- **Mastercard:** 5555 5555 5555 4444
- **Amex:** 3782 8224 6310 005

The system does not process real payments.

Q2: Why do authorization results sometimes fail?

The Beeceptor mock provider **randomly assigns outcomes** to simulate real-world conditions:

- Success
- Insufficient Funds
- Incorrect Details
- Server Error

Running multiple tests allows you to see each outcome.

Q3: Why does an order sometimes remain AUTHORIZED after settlement?

If the settlement amount is **less than the authorized amount**, the system treats this as a **partial settlement** and keeps the order in AUTHORIZED status until the remaining balance is settled.

Q4: Why do I see “Order Not Found” in the Settlement UI?

Most common causes:

- The Order ID was typed incorrectly
- The order has no **SUCCESS** authorization (failed authorizations cannot be settled)

Q5: Why do my changes not show up in the Order History UI?

This usually happens when the backend is restarted but the frontend was not refreshed. Reload the browser and try again.

Q6: How do I reset the system for a fresh demo?

1. Stop the backend
2. Restart backend
3. Restart frontend if needed

The system should start fresh.

Q7: Why does the checkout form format my card number automatically?

This is intended behavior.

The system uses:

- Card brand detection
 - Auto-formatting
 - Luhn checksum validation
 - Brand-aware CVV rules
- to simulate a real payment form.

Q8: Does the system store full card numbers or CVVs?

No.

For security, the system **only stores the last four digits** of the card number.

Full card numbers and CVVs are never saved.

Q9: What happens if Beeceptor goes offline?

The system will display a **Server Error** message.

Usually Beeceptor recovers within a minute or two.

Q10: Can orders be modified after authorization?

No.

To maintain workflow integrity:

- Authorization details cannot be changed
- Settlements cannot exceed the authorized limit
- Status transitions cannot be reversed

Appendix A: System Feature Overview

Below is a concise, user-friendly feature list for quick reference.

A.1 Checkout / Payment Processing Features

- Automatic Order ID & Amount generation
- Customer information capture
- Card brand detection (Visa, MasterCard, Amex, Discover)
- Auto-formatting of card numbers
- Luhn checksum validation
- Brand-aware CVV validation
- Expiration date validation
- Masking of sensitive card fields
- Real-time validation feedback
- Four possible authorization outcomes
- Result banners for user feedback
- Authorization saved in SQLite database

A.2 Warehouse Settlement Features

- Order lookup by ID
- Detailed order & authorization info panel
- Settlement amount validation
- Hard limit enforcement (cannot exceed authorization)
- Partial settlement support
- Automatic final settlement detection
- Status updates (AUTHORIZED → SETTLED)
- Settlement history tracking
- Automatic UI refresh after settlement
- Error and success banners

A.3 Order History / Reporting Features

- Current Orders tab
- Past Orders tab
- Status badges
- Search bar (order ID, customer name, status)
- Date range filtering

- Status filtering
- Column sorting (Order #, Customer, Date, Total, Status)
- Client-side pagination (20 / 50 / 100 rows per page)
- Order detail panel
- Authorization summary
- Settlement history
- CSV export (full filtered dataset)
- Mock fallback data if backend is offline

A.4 System-Level Features

- Node.js + Express backend
- SQLite database with foreign keys
- Beeeceptor mock payment provider integration
- API endpoints for authorization, settlement, and retrieval
- Automatic database initialization
- System reset by deleting database.db
- Fully local execution for demos

AI Use Disclaimer

Portions of this User Manual were developed with the assistance of artificial intelligence tools (OpenAI's ChatGPT). AI was used only to help draft, edit, and organize written instructions. All screenshots, system behavior descriptions, and procedural steps accurately reflect the functionality of the system as built by Group 10. The final content was reviewed and approved by the team.